

Database Methods 5N0783

Project Brief | 50% of module grade

Module	Database Methods 5N0783	Weighting	50%
Teacher	Joshua Aaron	Submission	[Date TBC]
Technique	Project	Submit via	Moodle (.ipynb + export + data)

Authorship Declaration: I hereby declare that all work described in this project, except where referenced, is entirely my own. All sources of information have been acknowledged by means of reference as required.

Overview

This project gives you the opportunity to choose a real-world dataset that genuinely interests you and to explore it using the database and data analysis tools we have covered in class — SQLite, pandas, and matplotlib. The project is structured as a Jupyter notebook that you will build up section by section, working at your own pace over the course of several weeks.

Each section has a clear purpose, and together they tell the story of your dataset: where it came from, what it contains, what questions you asked of it, and what you found. There is no single right answer here — the quality of your thinking and your explanations matters as much as the code itself.

If you have questions at any point, bring them to class or get in touch directly. Citing your sources is important throughout — if you are unsure how to do that, ask and we will go through it together.

What to Submit

Submit the following files via Moodle. Email submissions will not be accepted. If you have any difficulty with Moodle, get in touch and we will sort it out.

1. Your completed notebook as a **.ipynb** file (a copy of this one with all sections completed)
2. An **HTML or PDF export** of your notebook (File → Download as... or equivalent)
3. Your **data file(s)** in an appropriate format (.csv, .xlsx, etc.) so the notebook can be re-run. Use relative file paths. If files are too large for Moodle, provide a download link.
4. Any **images or additional files** referenced in the notebook
5. A separate **bibliography file** is acceptable if you are using a reference manager such as Zotero or Mendeley

Learning Outcomes Assessed

1	Outline a range of typical uses for databases
2	Describe essential database concepts including database files, tables, records, fields, queries and reports
3	Explain the purpose of query views such as design, datasheet, and SQL

9	Develop a database design including tables, keys and table relationships to meet a given design brief
10	Create a database which includes tables, queries, forms and reports based on a given design brief
11	Improve personal performance by using additional resources to solve familiar and unfamiliar database problems

Project Sections

The notebook is structured into nine numbered sections. You are free to add cells, move sections around, and adjust the structure to suit your dataset and your approach. If you are spending more time on one section than another, that is completely fine. The sections below describe what each part of the project involves.

Section 0	Introduction to Databases
	Open with a brief introduction that explains what databases are, where they are used in everyday life, and a short overview of what you will be doing throughout this project. This section sets the scene for your reader and gives context for the work that follows.
Section 1	Your Dataset
	Find at least one dataset on <i>Our World in Data</i> (ourworldindata.org) with at least four fields, on a topic that genuinely interests you. Download the data or access it via their API. Introduce your dataset here: what is it about, why did you choose it, and what questions might you want to explore?
Section 2	Importing Your Dataset
	Import your data in two ways: once using SQL commands directly in Python (via sqlite3 and pandas.to_sql), and once using pandas directly from a file or the API. Explain each approach and what it does.
Section 3	Describing Your Database
	Describe the key fields and features of your dataset. Which fields will you focus on for the rest of the project, and why? Include details such as data types, what each column represents, and the overall shape and size of the data.
Section 4	Querying Your Database
	Write SQL queries to explore your data and save the results to appropriately named variables. Explain each query: what does it return, and what does that tell you? Demonstrate how to add a record to your database, and store the fields you will use for your visualisations.
Section 5	Data Oddities and Issues
	Highlight anything unusual or problematic in your data — missing values, unexpected formatting, inconsistencies, or fields that need cleaning before they can be used. Perform any necessary cleaning or transformation, and add derived fields where they can reasonably be calculated from existing ones.
Section 6	Visualising Your Data
	Create charts and visualisations using Matplotlib , Seaborn , or Plotly (or a combination). Label all plots clearly with titles, axis labels, and legends where appropriate. Demonstrate how to export a visualisation as an image file.

Section 7	Reporting and Exporting
	Show how to create an exported report from your notebook (PDF, HTML, or a set of image files) and how to build at least one interactive visualisation using ipywidgets that allows the user to explore a part of your data. Include your exported report in your submission. Cite any sources you use.
Section 8	Conclusions and Further Investigation
	Reflect on what you found. What did the data tell you? What surprised you? Suggest ideas for further investigation and point your reader to other useful resources on the topic.
Section 9	Bibliography
	List all sources you used or cited throughout the project. If you used an LLM for assistance, include links to your conversation (with viewing permissions enabled) and screenshots of your prompts. All prompts should follow the Context / Tone / Format scheme discussed in class.

Assessment Criteria

Marks are awarded for objective, demonstrable evidence across four broad areas, matching the QQI marking categories for this component. If you are unsure whether something is sufficient, ask.

Discussion	5
Uses of databases described; essential concepts (tables, records, fields, queries) explained; purpose of query views outlined. (Sections 0, 3)	5
Design	20
Dataset selected from Our World in Data; topic introduced with reasons for choice; fields described with data types and focus identified. (Sections 1, 3)	10
Data oddities and issues identified; cleaning and/or derived fields applied and explained. (Section 5)	5
Conclusions drawn from the data; directions for further investigation suggested with supporting resources. (Section 8)	5
Implementation	20
Data imported via sqlite3/to_sql and via pandas; both approaches present and explained. (Section 2)	5
SQL queries written, saved to variables, and explained; INSERT demonstrated with conn.commit(); query results stored for visualisation. (Section 4)	5
At least two visualisations created, labelled with titles and axis labels, and exported as image files. (Section 6)	5
At least one ipywidgets interactive element created; notebook exported as HTML or PDF and included in submission. (Section 7)	5
Documentation	5
Bibliography complete with credible sources; LLM use documented with links and screenshots where applicable. (Section 9)	5
TOTAL	50